

Integral Partial Fraction

- Partial Fraction decomposition only gets applied when you have a rational function before you began applying decomposition you must make sure that the **denominators highest degree is bigger than the numerator highest degree**

$$\text{Rational Function} = \frac{f(x)}{g(x)} \text{ "Fraction"}$$

these are all examples of Partial Fraction and how to decompose this does not show how to fully solve for that you need to go a page or two down

example # 1

- You could not have this by u-sub or any other easy way

$$\int \frac{1}{(x-1)(x-2)}$$

$$\int \frac{a}{(x-1)} + \int \frac{b}{(x-2)}$$

- this is an example when to apply partial fraction

- this is decomposition you break it up by the power we had a power of one for each parentheses which was a total of two so we have two terms each with a different denominator

example # 2

- look at the denominator highest degree "x²" and look at the numerator highest degree "x" the denominator degree is larger which is what we want

$$\int \frac{5x-3}{x^2-3x-4}$$

$$\int \frac{a}{(x-4)} + \int \frac{b}{(x+1)}$$

- another example of a function where you need to apply partial fraction but you must first factor the denominator

- after factoring the denominator you can apply the decomposition

example #3

- the highest power in the denominator is 2 so in the decomposition we need to have two terms each term needs to have a different denominator
- a fraction is a whole term

$$\int \frac{7x+11}{(x-2)^2}$$

- another example where you need to apply partial fraction but this one has the same parentheses squared

$$\int \frac{a}{(x-2)} + \int \frac{b}{(x-2)^2}$$

- the amount of factors in the denominator for the decomposition must be the same amount as the highest power in the denominator of the original

example #4

- this has three terms in the denominator so when you decompose it you need three terms each will have a different denominator

$$\int \frac{x-4}{x(x-3)^2}$$

- this is another example where you would need to use partial fractions but this has a " $x(x-3)^2$ " so this would have a power of 3 in the denominator x and $(x+3)^2$

$$\int \frac{a}{x} + \int \frac{b}{(x-3)} + \int \frac{c}{(x-3)^2}$$

- this needs to have three terms in the decomposition because it has a power of three in the original denominator

example #5

- when ever you are decomposing a function that has quadratic in the denominator you need to give that quadratic $ax+b$ as a numerator and treat the rest accordingly

$$\int \frac{6x-1}{(x^2+3)(x-1)}$$

$$\int \frac{ax+b}{(x^2+3)} + \int \frac{c}{(x-1)}$$

- this is also another example where you would need to use partial fractions but this has a quadratic in the denominator
- you need to use $ax+b$ for a quadratic

example #6

$$\int \frac{x^2 + 3x + 5}{x+1}$$

- this is example where you need to apply Partial Fractions but this requires long division

$$\text{denominator} \frac{\overset{Q}{\text{numerator}}}{r}$$

$\Rightarrow$

$$\int \left(Q + \frac{r}{d} \right)$$

- after you finish apply long division you will place what you have into the equation (d stands for denominator from the original function)

how to solve

- after decomposing it you need to solve the integral

$$\int \frac{1}{(x-1)(x-2)} \Rightarrow \int \frac{a}{(x-1)} dx + \int \frac{b}{(x-2)} dx$$

$$a \int \frac{1}{u} du + b \int \frac{1}{u} du \Rightarrow a \ln|x-1| + b \ln|x-2|$$

- Solve for "a" and "b"

$$\left[\frac{1}{(x-1)(x-2)} = \frac{a}{(x-1)} + \frac{b}{(x-2)} \right] \quad \begin{array}{l} \text{(denominator)} \\ \bullet (x-1)(x-2) \Rightarrow 1 = a(x-2) + b(x-1) \end{array}$$

$$\begin{aligned} 1 &= a(x-2) + b(x-1) \\ x @ 1 \Rightarrow 1 &= a(1-2) + b(1-1) \\ 1 &= -a \Rightarrow \boxed{a = -1} \end{aligned}$$

- x can be anything the whole point is we want to cancel one of the letter while we keep the other we cancel "b" by (1-1) which is zero multiplied by "b" that left us with "a"

$$\begin{aligned}
 | &= a(x-2) + b(x-1) \\
 x @ 0 \Rightarrow | &= -1(0-2) + b(0-1) \\
 | &= 2 - b \\
 -1 &= -b \Rightarrow \boxed{b = 1}
 \end{aligned}$$

- We already solved for "a" we can plug it in and just solve for "b"

- We now have solved for "a" and "b" we can plug them in the answer we got from integrating

$$a \ln|x-1| + b \ln|x-2| \Rightarrow \boxed{-1 \ln|x-1| + 1 \ln|x-2|}$$

Practice Problems

#1) $\int \frac{5x+3}{(x-1)^2} dx$

$$\begin{aligned}
 \int \frac{a}{(x-1)} dx + \int \frac{b}{(x-1)^2} dx \quad u = x-1, \quad du = dx \\
 a \int \frac{1}{u} du + b \int \frac{1}{u^2} du \Rightarrow a \ln|x-1| - \frac{b}{(x-1)}
 \end{aligned}$$

$$\left[\frac{5x+3}{(x-1)^2} = \frac{a}{(x-1)} + \frac{b}{(x-1)^2} \right] \cdot (x-1)^2$$

$$5x+3 = a(x-1) + b$$

$$\begin{aligned}
 x @ 1 \Rightarrow 5(1)+3 &= a(1-1) + b \\
 \boxed{8} &= b
 \end{aligned}$$

$$\begin{aligned}
 x @ 0 \Rightarrow 5(0)+3 &= a(0-1) + 8 \\
 3 &= -a + 8 \\
 -5 &= -a \Rightarrow \boxed{a = 5}
 \end{aligned}$$

$$a \ln|x-1| - \frac{b}{(x-1)}$$

$$\boxed{5 \ln|x-1| - \frac{8}{(x-1)}}$$

#2) $\int \frac{x+1}{x+2} dx$

$$\int \frac{x+1+1-1}{x+2} dx \Rightarrow \int \frac{x+2}{x+2} dx - \int \frac{1}{x+2} dx$$

$$\int 1 dx - \int \frac{1}{x+2} dx \Rightarrow x - \int \frac{1}{u} du \quad u = x+2, \quad du = dx$$

$$\boxed{x - \ln|x+2|}$$

- this is a very trick one but notice that we have not changed anything even though we add terms to it

$$\#3) \int \frac{3x+4}{(x^2+1)(x-2)} dx$$

$$\int \frac{ax+b}{(x^2+1)} dx + \int \frac{c}{(x-2)} dx$$

$$\int \frac{ax}{(x^2+1)} dx + \int \frac{b}{(x^2+1)} dx + \int \frac{c}{(x-2)} dx$$

$$u = x^2 + 1 \quad u = x - 2$$

$$du = 2x \cdot dx \quad du = dx$$

$$\frac{du}{2x} = dx$$

$$a \int \frac{x}{u} \cdot \frac{du}{2x} + b \int \frac{1}{x^2+1} dx + c \int \frac{1}{u} \cdot dx$$

$$\frac{a}{2} \cdot \ln|x^2+1| + b \tan^{-1}(x) + c \ln|x-2|$$

$$\left[\frac{3x+4}{(x^2+1)(x-2)} = \frac{ax}{(x^2+1)} + \frac{b}{(x^2+1)} + \frac{c}{(x-2)} \right] \cdot (x^2+1)(x-2)$$

$$3x+4 = ax(x-2) + b(x-2) + c(x^2+1)$$

$$x @ 2 \Rightarrow 3(2)+4 = a(2)(2-2) + b(2-2) + c(2^2+1)$$

$$10 = 5c \Rightarrow \boxed{c=2}$$

$$x @ 0 \Rightarrow 3(0)+4 = a(0)(0-2) + b(0-2) + c(0^2+1)$$

$$4 = -2b + 2$$

$$2 = -2b \Rightarrow \boxed{b=-1}$$

$$x @ 1 \Rightarrow 3(1)+4 = a(1)(1-2) + b(1-2) + c(1^2+1)$$

$$7 = -a - 1 + 4$$

$$7 = -a + 3$$

$$4 = -a \Rightarrow \boxed{a=-4}$$

$$\frac{a}{2} \cdot \ln|x^2+1| + b \tan^{-1}(x) + c \ln|x-2|$$

$$\boxed{-2 \ln|x^2+1| - \tan^{-1}(x) + 2 \ln|x-2|}$$

$$\#4) \int \frac{x^3+1}{x^2-1} \quad x^2-1 \sqrt{\frac{x^3+1}{x^2-1}} = \frac{x^3+1}{x^2-1}$$

$$\int \left(q + \frac{r}{d} \right) = \int x + \frac{x+1}{x^2-1}$$

$$\int x + \int \frac{x+1}{(x-1)(x+1)} = \int \frac{a}{x-1} + \int \frac{b}{x+1}$$

$$u = x-1 \quad u = x+1$$

$$du = dx \quad du = dx$$

$$a \int \frac{1}{u} + b \int \frac{1}{u} = a \ln|x-1| + b \ln|x+1|$$

$$\left[\frac{x+1}{(x+1)(x-1)} = \frac{a}{(x-1)} + \frac{b}{(x+1)} \right] \cdot (x-1)(x+1)$$

$$x+1 = a(x+1) + b(x-1)$$

$$x @ 1 \Rightarrow 2 = a(1+1) + b(1-1)$$

$$2 = 2a \Rightarrow \boxed{a=1}$$

$$x @ -1 \Rightarrow 0 = 1(-1+1) + b(-1-1)$$

$$0 = -2b \Rightarrow \boxed{b=0}$$

$$\int x + a \ln|x-1| + b \ln|x+1|$$

$$\boxed{\frac{x^2}{2} + \ln|x-1|}$$

review on long division

$$\frac{x^2 + 3x + 2}{x + 1}$$

step 1

$$x+1 \overline{) x^2 + 3x + 2}$$

$$\frac{x}{x} \quad \text{(Place it on top)}$$

- Step 1 we divide the first term of the Numerator and the first term of denominator and place that amount at the top

step 2

$$x \cdot (x + 1) \\ (x^2 + x)$$

- step 2 we take that "x" from the top and multiply it with the denominator

step 3

$$x+1 \overline{) x^2 + 3x + 2} \\ - (x^2 + x) \quad \downarrow \\ \hline 2x + 2$$

- Step 3 we take the result from distribution and subtract it from the Numerator when subtracting it looks like this: $(x^2 + 3x + 2) - (x^2 + x)$ don't forget to distribute the negative

step 4

$$x+1 \overline{) x^2 + 3x + 2} \\ - (x^2 + x) \quad \downarrow \\ \hline 2x + 2 \\ - (2x + 2) \\ \hline 0$$

Quotient (Q) ←

← remainder (r)

- Step 4 we do the same thing all over we divide by the first term of the Numerator and the first term of denominator and place that amount at the top and distribute it into the denominator and subtract the result from the numerator

$$Q + \frac{r}{d} = x + 2 + \frac{0}{x+1} \Rightarrow \boxed{x + 2}$$

$$\frac{x^2 + 3x + 2}{x + 1} \quad \leftarrow \text{denominator (d)}$$